

# **Supplemental Material**

## **Data S1.**

### **Supplemental Methods**

#### *Cardiovascular Event Subgroups*

For AMI, we examined ST-segment elevation MI (STEMI) (ICD-9: 410.0x-410.6x, 410.8x; ICD-10: I21.0x-I21.2x, I21.3, I22.0-I22.1, I22.8-I22.9) and non-ST-segment elevation MI (NSTEMI) (ICD-9: 410.7x, 410.9x; ICD-10: I21.4 and I22.2). For stroke, we examined ischemic stroke (ICD-9: 433.x1; 434.x1; 436.x; ICD-10: I63.x) and hemorrhagic stroke (ICD-9: 430.x-432.x; ICD-10: I60.x-I62.x).

### **Supplemental Results**

#### *Overall Associations for Event Subgroups*

Results for subgroups of AMI showed protective effects, with increasing PM<sub>2.5</sub> exposure associated with larger decreased risk of STEMI AMI [hazard ratio 0.72 (95% CI: 0.55, 0.95)] than NSTEMI AMI [hazard ratio 0.89 (95% CI: 0.79-0.99)]. Results for subgroups of stroke were not statistically significant: relative risk of ischemic stroke was 0.92 (95% CI: 0.81-1.06) and relative risk of hemorrhagic stroke was 0.93 (95% CI: 0.70-1.25).

#### *Extended Models*

In an extended fully-adjusted model using median household income instead of education as a measure of neighborhood SES, the hazard ratios were 1.22 (95% CI: 1.13, 1.32), 1.36 (95% CI: 1.23, 1.50), 0.77 (95% CI: 0.61, 0.96), 0.92 (95% CI: 0.81, 1.04), and 0.87 (0.79, 0.97) for CVD mortality, IHD mortality, cerebrovascular mortality, stroke, and AMI, respectively, revealing no significant difference in results when compared to models using neighborhood education.

### *Effect Modification*

We looked at effect modification by HTN medication use and statin medication use but found no statistically significant interactions: HTN medication interaction p-value=0.767 and statin medication interaction p-value=0.830. In **Figure S2**, one race subgroup (American Indian/Alaska Native) had a very low relative risk of 0.41 (95% CI: 0.10-1.64), yet there were only 30 IHD mortality events in this group, which contributes to the wide confidence interval observed.

**Table S1. Cardiovascular (CVD) mortality event subtypes.**

| <b>CVD Mortality (ICD-10: I10.x–I70.x)</b>                  |                                      | <b>N = 13,385 events</b> |
|-------------------------------------------------------------|--------------------------------------|--------------------------|
| <b>Event Subtype (ICD-10 code)</b>                          | <b>N (% of CVD Mortality events)</b> |                          |
| IHD Mortality (I20.x-I25.x)                                 | 7,563 (56.5%)                        |                          |
| Cerebrovascular Mortality (I60.x-I69.x)                     | 1,718 (12.8%)                        |                          |
| Heart Failure (I50.x)                                       | 1,126 (8.4%)                         |                          |
| Hypertensive Diseases (I10.x-I16.x)                         | 1,026 (7.7%)                         |                          |
| Other CVD Mortality (I26.x-I28.x, I30.x-I49.x, I51.x-I52.x) | 1,952 (14.6%)                        |                          |

**Table S2. Ischemic heart disease (IHD) mortality event subtypes.**

| <b>IHD Mortality (ICD-10: I20.x-I25.x)</b>                       |                                      | <b>N = 7,563 events</b> |
|------------------------------------------------------------------|--------------------------------------|-------------------------|
| <b>Event Subtype (ICD-10 code)</b>                               | <b>N (% of IHD Mortality events)</b> |                         |
| Acute myocardial infarction (I21.x)                              | 2,310 (30.5%)                        |                         |
| Acute myocardial infarction, unspecified (I21.9)*                | 2,290 (30.3%)                        |                         |
| Chronic ischemic heart disease (I25.x)                           | 5,185 (68.6%)                        |                         |
| Atherosclerotic cardiovascular disease, so described (I25.0)*    | 399 (5.3%)                           |                         |
| Atherosclerotic heart disease of native coronary artery (I25.1)* | 4,318 (57.1%)                        |                         |
| Ischemic cardiomyopathy (I25.5)*                                 | 254 (3.4%)                           |                         |
| Other IHD Mortality (I20.x, I22.x, I23.x, I24.x)                 | 68 (0.9%)                            |                         |

\*Most common ICD subcodes of IHD event subtypes in this cohort

**Table S3. Recurrent and incident acute myocardial infarction (AMI) and stroke events by cardiovascular disease history group.**

| <b>Group</b>                          | <b>N</b> | <b>AMI events in follow-up</b>                                                    | <b>Stroke events in follow-up</b>                                                          |
|---------------------------------------|----------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>History of AMI Only</b>            | 31,926   | 3,229 Recurrent AMI events                                                        | 1,278 Incident stroke events                                                               |
| <b>History of Stroke Only</b>         | 54,723   | 3,086 Incident AMI events                                                         | 3,645 Recurrent stroke events                                                              |
| <b>History of Both AMI and Stroke</b> | 9,933    | 1,367 Recurrent AMI events                                                        | 742 Recurrent stroke events                                                                |
| <b>All</b>                            | 96,582   | 7,682 Total AMI events<br>3,086 Incident AMI events<br>4,596 Recurrent AMI events | 5,665 Total stroke events<br>1,278 Incident stroke events<br>4,387 Recurrent stroke events |

**Table S4. Risk of cardiovascular events associated with an exposure increase of 10 µg/m<sup>3</sup> in the level of time-varying annual mean PM<sub>2.5</sub> exposure among patients with a history of acute myocardial infarction (AMI) and/or stroke.**

| Outcome                                               | Number of events (% of cohort N=96,582) | Model 1*<br>Hazard Ratio (95% CI) | Model 2†<br>Hazard Ratio (95% CI) | Model 3‡<br>Hazard Ratio (95% CI) | Model 4§<br>Hazard Ratio (95% CI) |
|-------------------------------------------------------|-----------------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| <b>Acute Myocardial Infarction (Mostly non-fatal)</b> | 7,682 (7.95)                            | 1.00 (0.91-1.11)                  | 0.92 (0.83-1.02)                  | 0.94 (0.85-1.04)                  | 0.86 (0.77-0.96)                  |
| <b>Stroke</b>                                         | 5,663 (5.86)                            | 0.95 (0.84-1.07)                  | 0.93 (0.83-1.05)                  | 0.93 (0.82-1.05)                  | 0.93 (0.82-1.05)                  |
| <b>CVD Mortality</b>                                  | 13,385 (13.86)                          | 1.35 (1.25-1.45)                  | 1.31 (1.21-1.41)                  | 1.29 (1.20-1.39)                  | 1.20 (1.11-1.30)                  |
| <b>IHD Mortality</b>                                  | 7,563 (7.83)                            | 1.54 (1.39-1.70)                  | 1.45 (1.32-1.60)                  | 1.45 (1.31-1.60)                  | 1.34 (1.21-1.48)                  |
| <b>Cerebrovascular Mortality</b>                      | 1,718 (1.78)                            | 0.76 (0.61-0.95)                  | 0.81 (0.65-1.01)                  | 0.80 (0.64-1.00)                  | 0.73 (0.58-0.92)                  |

\*Model 1: Sex, race/ethnicity, study start year (3-yr groups)

†Model 2: Basic Model + comorbidities (hypertension, hyperlipidemia, diabetes) + BMI + smoking + CVD history

‡Model 3: Model 2 + revascularization (CABG and PCI) + medications (hypertension meds and statin meds)

§Model 4: Model 3 + SES (neighborhood education + Medicaid insurance)

||All models with baseline hazards by 5-year age categories

**Table S5. Sensitivity analyses using fully adjusted model with baseline PM<sub>2.5</sub> exposure.**

| <b>Analyses</b>                       | <b>CVD Mortality HR<br/>(95% CI)</b> | <b>Stroke HR (95% CI)</b> | <b>AMI HR (95% CI)</b> |
|---------------------------------------|--------------------------------------|---------------------------|------------------------|
| <b>Baseline analyses</b>              | 1.23 (1.13-1.33)                     | 0.92 (0.80-1.04)          | 0.91 (0.81-1.01)       |
| <b>Age as time scale</b>              | 1.23 (1.13-1.34)                     | 0.92 (0.81-1.05)          | 0.91 (0.81-1.02)       |
| <b>Strata by study start<br/>year</b> | 1.22 (1.12-1.33)                     | 0.92 (0.80-1.04)          | 0.91 (0.81-1.01)       |
| <b>Death as competing risk</b>        | 1.17 (1.07-1.27)                     | 0.87 (0.76-0.99)          | 0.87 (0.78-0.98)       |

\*Hazard ratios from fully-adjusted model using PM<sub>2.5</sub> exposure at baseline

**Table S6. Risk of cardiovascular events associated with an exposure increase of 10 µg/m<sup>3</sup> in the level of time-varying annual mean PM<sub>2.5</sub> exposure among patients with a history of stroke, stratified by time since prior stroke event.**

| <b>Outcome</b>       | <b><u>Previous stroke event &lt;5 years ago</u></b><br><b>N=9,709</b> |                    | <b><u>Previous stroke event ≥5 years ago</u></b><br><b>N=4,754</b> |                    | <b><u>Previous stroke event date unknown</u></b><br><b>N=82,119</b> |                    |
|----------------------|-----------------------------------------------------------------------|--------------------|--------------------------------------------------------------------|--------------------|---------------------------------------------------------------------|--------------------|
|                      | <b>Number of<br/>Events (%)</b>                                       | <b>HR (95% CI)</b> | <b>Number of<br/>Events (%)</b>                                    | <b>HR (95% CI)</b> | <b>Number of<br/>Events (%)</b>                                     | <b>HR (95% CI)</b> |
| <b>AMI</b>           | 532 (5.48)                                                            | 0.90 (0.60-1.36)   | 329 (6.92)                                                         | 1.17 (0.68-1.99)   | 6,821 (8.31)                                                        | 0.85 (0.76-0.96)   |
| <b>Stroke</b>        | 987 (10.17)                                                           | 0.90 (0.67-1.22)   | 419 (8.81)                                                         | 0.75 (0.46-1.23)   | 4,257 (5.18)                                                        | 0.95 (0.82-1.09)   |
| <b>CVD Mortality</b> | 1,570 (16.17)                                                         | 1.21 (0.96-1.53)   | 857 (18.03)                                                        | 1.33 (0.95-1.85)   | 10,958 (13.34)                                                      | 1.21 (1.11-1.32)   |

\*Hazard ratios from fully-adjusted model

**Table S7. Risk of cardiovascular events associated with an exposure increase of 10 µg/m<sup>3</sup> in the level of time-varying annual mean PM<sub>2.5</sub> exposure among patients with a history of acute myocardial infarction (AMI), stratified by time since prior AMI event.**

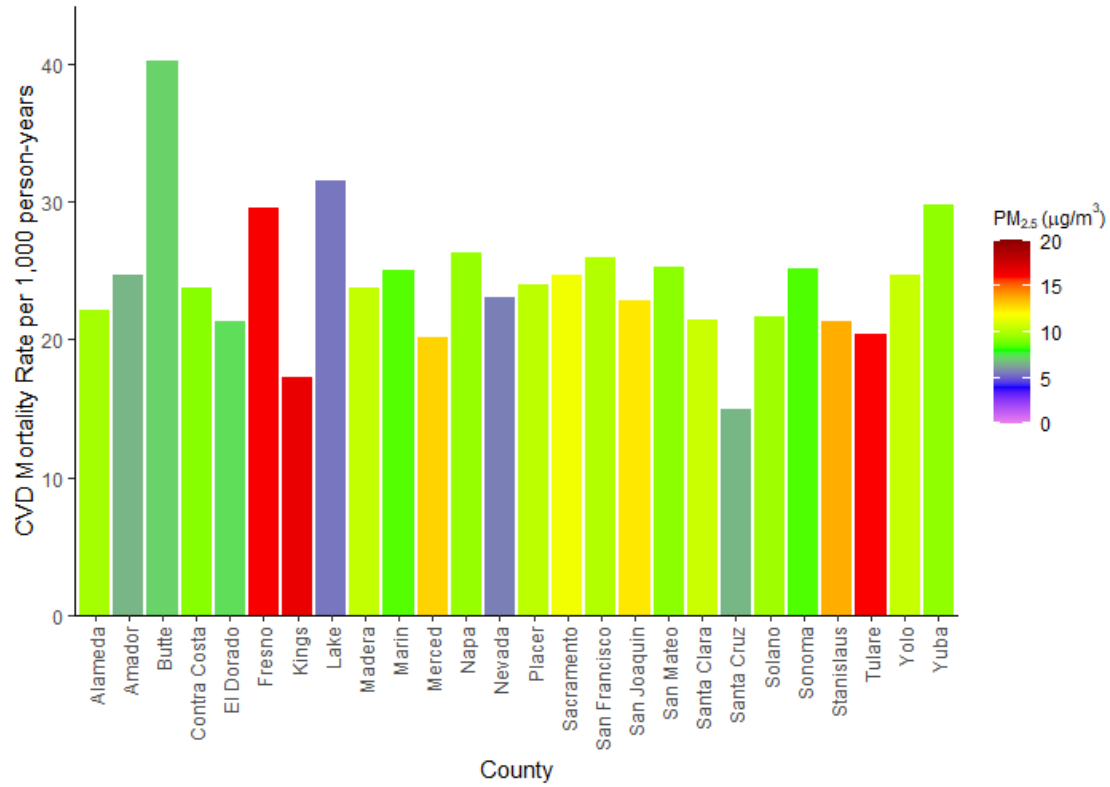
| Outcome       | <u>Previous AMI event &lt;5 years ago</u><br>N=18,014 |                  | <u>Previous AMI event ≥5 years ago</u><br>N=10,284 |                  | <u>Previous AMI event date unknown</u><br>N=68,284 |                  |
|---------------|-------------------------------------------------------|------------------|----------------------------------------------------|------------------|----------------------------------------------------|------------------|
|               | Number of Events (%)                                  | HR (95% CI)      | Number of Events (%)                               | HR (95% CI)      | Number of Events (%)                               | HR (95% CI)      |
| AMI           | 2,321 (12.88)                                         | 0.90 (0.74-1.10) | 1,148 (11.16)                                      | 1.05 (0.79-1.38) | 4,213 (6.17)                                       | 0.83 (0.72-0.96) |
| Stroke        | 841 (4.67)                                            | 0.87 (0.63-1.20) | 545 (5.30)                                         | 0.96 (0.64-1.44) | 4,277 (6.26)                                       | 0.92 (0.79-1.06) |
| CVD Mortality | 2,681 (14.88)                                         | 1.10 (0.93-1.31) | 1,561 (15.18)                                      | 1.37 (1.09-1.72) | 9,143 (13.39)                                      | 1.22 (1.11-1.34) |

\*Hazard ratios from fully-adjusted model

**Table S8. Race and neighborhood high school education comparison between cohort and 2010 Census.**

| <b>Demographic</b>                                | <b>Study Cohort</b> | <b>California</b> | <b>United States Overall</b> |
|---------------------------------------------------|---------------------|-------------------|------------------------------|
| <b>White</b>                                      | 72%                 | 72%               | 76%                          |
| <b>Black</b>                                      | 8%                  | 7%                | 13%                          |
| <b>American Indian/Alaska Native</b>              | 1%                  | 2%                | 1%                           |
| <b>Asian/Pacific Islander</b>                     | 10%                 | 16%               | 6%                           |
| <b>Multiple Races</b>                             | 8%                  | 4%                | 3%                           |
| <b>Hispanic/Latino (Any race)</b>                 | 12%                 | 39%               | 19%                          |
| <b>Percent with less than high school diploma</b> | 14%                 | 17%               | 12%                          |

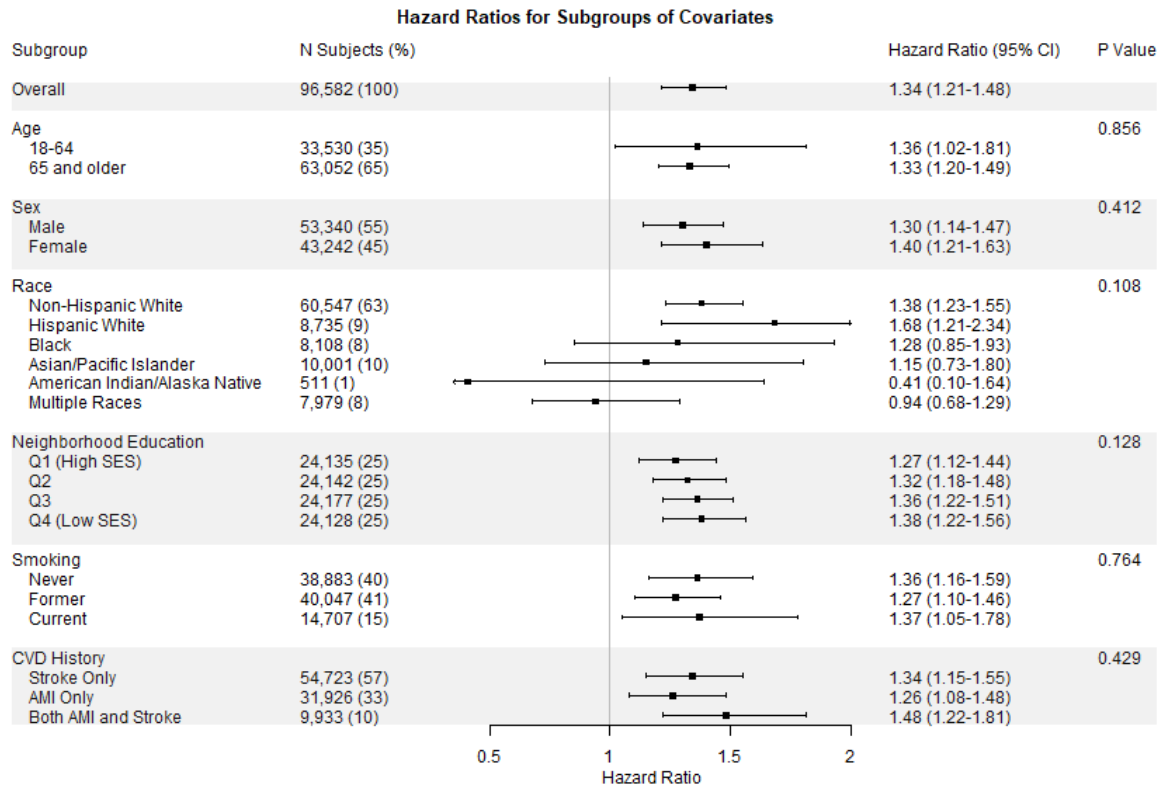
**Figure S1. Cardiovascular mortality rate and Kaiser Permanente Northern California (KPNC) county means of 1-year average PM<sub>2.5</sub> exposure among study subjects at baseline.**



\*CVD mortality rate per 1,000 person-years = (# of CVD deaths/total person-years of follow-up) x 1000

†Only counties with >10 CVD mortality cases are displayed in graph

**Figure S2. Risk of ischemic heart disease (IHD) mortality associated with an exposure increase of 10  $\mu\text{g}/\text{m}^3$  in the level of time-varying annual mean  $\text{PM}_{2.5}$  exposure among subjects with a history of acute myocardial infarction (AMI) and/or stroke by subgroup.**



\*Hazard ratios from fully-adjusted model

†P-values for interaction

‡Education p-value: interaction effect as a trend